

Nikhil Mandayam Adyapak

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Seeking Full-Time Roles in ML Engineering, MLOps, Software Engineering, Data Science, or Data Engineering.
Available June 2027.

EDUCATION

University of Wisconsin-Madison

Sep 2025 – May 2027

Masters in Data Science

GPA - 3.68/4.00

Coursework: Machine Learning, Foundation Models, Database Design & Implementation, Statistical Models, Methods, Inference for Data Science | **Teaching Assistantship:** CS564 Database Management Systems

PES University, Bengaluru, India

Aug 2019 – May 2023

Bachelor of Technology in Computer Science & Engineering

GPA - 3.78/4.00

Specialization: Machine Intelligence & Data Science | **Teaching Assistantship:** CS203 Statistics for Data Science

Coursework: Algorithms, Data Structures, Data Science, Linear Algebra, Machine Learning, Big Data, Cloud Computing

WORK EXPERIENCE

Ciroos

SF Bay Area

AI Intern

May 2026 – Dec 2026

- Productionized **Agentic Search** over incident chats for SRE agent context from incident history.
- Built a **Graph RAG** context engine on a temporal knowledge-graph that sources SRE agent incident answers from team chat and investigation history.
- Prototyped a graph-free **vector RAG** alternative over incident chats.
- Evaluated SRE agents by simulating faults across **Kubernetes** (AWS, GCP, Azure); **RCA** instruction for agents.

Bosch Global Software Technologies (BGSW)

Bengaluru, India

Senior Software Engineer, ADAS

Aug 2023 – Aug 2025

MLOps (ML in Operations) & Distributed Cloud Workflows

- Delivered an **MLOps platform** for pedestrian detection (**Azure ML Studio** + **MLFlow**) with end-to-end ML lifecycle; enabled **multi-GPU training** (3x faster) and cut GPU costs by **60%**.
- Scaled a **Ray Cluster** based distributed system for **ML training** on Azure Kubernetes Service (**AKS**), reducing retraining cycles from **4 weeks** to **1 week (75% faster)** and supporting **30+** teams.
- Automated platform CI/CD and cloud automation on **AKS** (Terraform, Helm, Kustomize, **Argo**, GitHub Actions); integrated **Prometheus** observability and **Alertmanager**, accelerating experiment setup for **multi-tenant** ADAS teams from **3 days** to **5 minutes**.
- Deployed an on-prem **Ollama LLM** as a Python **FastAPI backend microservice**, enabling private, team-wide LLM access (**150+ users**), zero external data egress.

Core Machine Learning & Model Evaluation

- Enhanced ADAS image dataset resolution by **4x** using **ESRGAN** transfer learning in **PyTorch** on curated datasets.
- Benchmarked **occlusion estimation** models on nuScenes dataset with a LiDAR-depth evaluation pipeline (accuracy, confusion matrices, F1, visualizations).
- Researched cost-aware labeling by image complexity; cut labeling effort and ground-truth cost by about **30%**.

Large-Scale Image Retrieval & Dataset Generation for Autonomous Driving

- Architected a containerized **FastAPI backend microservice** for a **60M+** vector image-search engine, deploying CLIP embeddings and ElasticSearch to reduce edge-case triage by **95%** (4 hours to 10 mins).
- Engineered a **scene-understanding** image retrieval pipeline using foundation models (Mask2Former, Depth-Anything, OWL-ViT, **CLIP**) to generate scene graphs, curating fine-grained datasets for ADAS corner cases.
- Integrated **CLIP embedding** re-use with **80%** cross-version alignment, avoiding full re-indexing of legacy image vectors.

Bosch Global Software Technologies (BGSW)

Bengaluru, India

MLOps Intern

Jan 2023 – May 2023

- Established on-premise **MLOps** pipelines (**DVC**, **MLflow**) and accelerated **multi-GPU** object detection training (**YOLOv5**, **Detectron2**) in **PyTorch** on **Azure ML Studio**, cutting setup time from **1 week** to **1 day (85% faster)** and training time by **67%**.
- Containerized model **microservices** with **Docker**; **Azure DevOps** CI/CD to **Azure Container Registry**.

- Designed an **MLOps** pipeline for object detection and segmentation (Detectron2, YOLOv5) in **PyTorch** with **DVC**.
- Implemented **ensemble strategies** and performance visualizations for autonomous driving.

TECHNICAL SKILLS

Programming languages: Python, SQL, C, Java, R, C++

ML & Statistics: PyTorch, TensorFlow, HuggingFace, Transformers, CLIP, LLM, CNN, GAN, Scikit-learn, OpenCV, Pandas, NumPy, matplotlib

GenAI: RAG, LangChain, ChromaDB

MLOps & Workflows: Ray, MLFlow, DVC, Argo Workflows, Azure ML Studio, Prometheus, Grafana, Alertmanager

Cloud: Azure, AWS, GCP, AKS, Docker, Terraform, Helm, Azure DevOps, GitHub Actions, Linux, Shell

Data: ETL pipelines, PySpark, Spark, Databricks, Kafka

Databases & Tools: Elasticsearch, PostgreSQL, MySQL, MongoDB, FastAPI, Flask, Streamlit, GitHub

PROJECTS AND PUBLICATIONS

[1] **Code Runtime Complexity Prediction** - ERCICA (Springer), 2023 | Python, TensorFlow, sklearn, Streamlit, NetworkX
Predicted Big-O runtime complexity of C/Java/Python code using static analysis and ML classification with BiLSTM over Abstract Syntax Trees graph embeddings on IBM CodeNet dataset (**Accuracy: 96%**). [[View Publication](#)]

[2] **Sentinel-Edge AI** - MadData Hackathon 2026 (Qualcomm Track) | Llama 3.2, RAG, FastAPI
Selected as **1 of 10 teams** to build an air-gapped **Edge AI** security auditor on Snapdragon X Elite NPU, with a RAG pipeline and ChromaDB and no cloud egress. [[View Project](#)]

[3] **MLOps POC pipeline for Pedestrian Detection** - 2023 | Python, PyTorch, DVC, MLFlow, Detectron2
Shipped an on-premise MLOps starter template with DVC and Detectron2 for the ML lifecycle. [[View Project](#)]

[4] **DistilBERT natural language inference** - PyTorch, HuggingFace, Transformers
Fine-tuned DistilBERT for natural language inference (**83%** test accuracy). [[View Project](#)]

[5] **Vision-language hallucination study** - PyTorch, CLIP, Transformers
LLaVA-1.5 hallucination check against the CLIP vision encoder. [[View Project](#)]

[6] **Novel ways of decrypting transposition ciphers** - IEEE SMARTGENCON, 2022 | Python
Introduced cipher-breaking optimization techniques over columnar transposition ciphers. [[View Publication](#)]